
Technical Procedure

Workflow · Methodology · Reproducibility

Team NeurAps · 5 September 2026

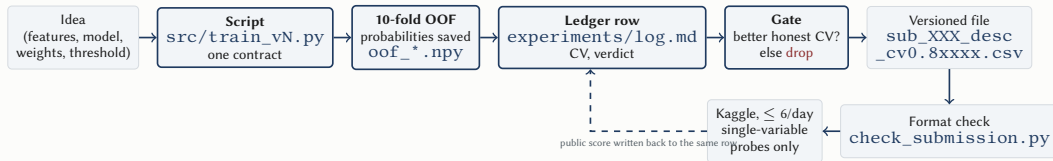
Everything shown here is a file in our repository or a cell in the submitted notebook. Ask us to open any of them.

Part 3 · Section 1

Workflow

1

The loop we ran, end to end, for every idea



Rules the loop enforced

- `data/raw/` is never modified; derived artifacts go to `data/processed/`.
- One shared `build_features()` is the single source of truth for every model.
- Every script prints per-model OOF, blend weights and the tuned threshold; nothing is judged by eye.
- A result is recorded before the next idea starts; drops are rows, not deletions.

Repository layout

- `src/`: one script per pipeline version, plus sweeps, tuning, stacking, checks
- `data/processed/`: the component bank — OOF and test probabilities for every model trained
- `submissions/`: 30 versioned files, 23 sent to the leaderboard
- `experiments/log.md`, `memory/insights.md`, `notebooks/`: ledger, findings, the reproducible solution

The ledger: what a row looks like

Every experiment, kept or dropped, is one row. Three real rows:

#	Change	CV (OOF weighted F1)	Public	Verdict
2	Engineered features + LGBM/XGB/CatBoost blend, threshold 0.59	0.87768 (lgb .87704 / xgb .87672 / cat .87721)	0.876023	keep — CV +0.0007 gave LB +0.0017
3	Optuna-tuned LGBM + XGB + Cat, 10-fold $\times$ 3 seeds, threshold 0.56	0.87964	0.875246	CV $\leftrightarrow$ LB diverged; probe the threshold next
17	RF/ET/MLP diversity: v4 + 5% each, threshold 0.575	0.87996 (split-half: .87934 vs .87876 $\Rightarrow$ real +0.0006)	0.875506	best honest CV — final pick candidate

Why the ledger mattered technically

The endgame question — “which two files?” — was answered by sorting this table on the honest-CV column, not by memory. The public column exists in the same row so that CV $\leftrightarrow$ LB gaps are visible per experiment.

Part 3 · Section 2

Methodology

2

Feature construction in code

Step	Implementation
Sentinel decoding	<code>pd.to_numeric(col, errors="coerce")</code> ; any value ≥ 990 set to missing (999 / 998 are “unknown” codes, “Blank(s)” and “Unknown or size unreasonable” become NaN)
Tumor size coalesce	<code>summary.fillna(overtime).fillna(cs_2004_2015)</code> + <code>size_missing_all</code> flag
Ordinal maps	T: T0/Tis $\rightarrow$ 0, T1mi/T1a $\rightarrow$ 1, T1b $\rightarrow$ 1.5, T1c $\rightarrow$ 2, T2a $\rightarrow$ 2.5, T2b $\rightarrow$ 3, T3 $\rightarrow$ 4, T4 $\rightarrow$ 5; N0–N3 $\rightarrow$ 0–3; M0 $\rightarrow$ 0, M1a $\rightarrow$ 1, M1b $\rightarrow$ 2, M1c $\rightarrow$ 3; grade I–IV $\rightarrow$ 1–4; stage L/R/D $\rightarrow$ 1–3
Composite	<code>tnm_sum = t + n + 2*m</code> , unknowns filled with scale midpoints (2.5, 1.5, 1.5)
Counts and flags	<code>mets_count</code> (Yes across bone/brain/liver/lung), <code>mets_known</code> , <code>surgery_done</code> , <code>death_cert_only</code> (regex on the reason column), <code>radiation_given</code> , <code>any_treatment</code> , <code>multiple primaries</code>
Ratios and time	<code>nodes_ratio = positive / max(examined, 1)</code> where <code>examined > 0</code> ; <code>years_since_dx = 2024 - year</code>
Categoricals	pandas category with levels unioned over train and test (<code>union_categoricals</code>) so every library sees identical codes
Three encodings	LightGBM: native categories · XGBoost/RF/ET: integer codes · CatBoost/MLP: strings with an explicit “NA” level

35 raw predictors $\rightarrow$ 53 model columns. No target statistics outside a fold; nothing fitted on test.

Validation protocol, as implemented

- `StratifiedKFold(10, shuffle=True, random_state=seed)`, seeds 42 / 2026 / 7.
- Each fold: train on 9 parts (+ pseudo-rows), early-stop on the 10th, predict the 10th $\Rightarrow$ one out-of-fold probability per training patient.
- OOF averaged over seeds; test probability averaged over all fold models $\times$ seeds. No refit on the full set: the fold models are early-stopped, and averaging them is a free ensemble.
- Threshold: sweep 0.35–0.75 in steps of 0.005 on pooled OOF. Report the global argmax *and* the median of per-fold optima; when they disagree, the score is fragile.

Split-half verification (pseudo-code)

```
A, B = random halves of OOF rows
choice = argmax over candidates of
score(A)
report score(B) for that frozen choice
compare with score(B) of the incumbent
```

What it decided

Diversity mixture: survived (+0.0006). Per-year thresholds: failed.
Distillation: never passed (mechanism leaked). Stacking: no gain.

Model configurations, exactly

Model	Parameters	Input encoding
LightGBM (both stages)	lr 0.05, num_leaves 27, min_data_in_leaf 20, feature_fraction 0.40, bagging_fraction 0.60 (freq 1), λ_1 7.36, λ_2 0.078, min_gain_to_split 0.19; $\leq$ 3000 rounds, early stopping 100	native categoricals
XGBoost	lr 0.05, max_depth 7, subsample 0.8, colsample_bytree 0.8, min_child_weight 10; log-loss early stopping 100	integer codes
CatBoost	lr 0.05, depth 7, l2_leaf_reg 5; Logloss early stopping 100; ordered target statistics	strings + “NA”
RandomForest / ExtraTrees	800 trees, min_samples_leaf 5, max_features $\sqrt{p}$	integer codes, median impute
MLP	layers 128→64 ReLU, α 1e-3, batch 512, $\leq$ 60 epochs, early stopping (patience 8)	one-hot (min freq 20) + standardised numerics
Optuna study	40 trials, TPE sampler, 5-fold; objective = mean per-fold best-threshold weighted F1; space: leaves 15–255, min_data 10–200, feature/bagging fractions, λ_1 , λ_2 , min_gain	LightGBM only

The tuned LightGBM parameters are hard-coded in the notebook and used in both stages. XGBoost and CatBoost were never tuned; our controlled probe showed no leaderboard gain from tuning, so no further budget went there.

Blend, mixture and decision threshold

GBM blend weights

- Objective: $-\max_t \text{wF1}(y, \sum_m w_m p_m > t)$ on OOF.
- Optimiser: Nelder–Mead (derivative-free simplex), because thresholded F1 is a step function.
- $w \leftarrow |w| / \sum |w|$; solutions landed near 1/3 each (e.g. 0.344 / 0.333 / 0.323).

Final probability

$$p = 0.85 p_{\text{GBM}} + 0.05 (p_{\text{RF}} + p_{\text{ET}} + p_{\text{MLP}})$$

Decision

- Sweep on pooled OOF $\Rightarrow$ 0.575; per-fold optima 0.570–0.575.
- Label Dead iff $p > 0.575$. Predicted Dead rate 85.8% (train 83.0%).
- OOF at 0.575: weighted F1 0.87996; at 0.5: 0.87518. Accuracy is 88.4% at both cuts — only the metric moves.

Duplicate correction

283 test rows are exact 35-column copies of train rows; they take the train label. Changed 1 prediction.

Pseudo-labelling, inside the fold

The augmentation, per fold

```
mask = (p_test > 0.99) | (p_test < 0.02)
X_ps, y_ps = X_te[mask], (p_test[mask] > .5)

for i_tr, i_va in folds:
    X_aug = concat(X[i_tr], X_ps)
    y_aug = concat(y[i_tr], y_ps)
    model.fit(X_aug, y_aug,
              eval_set=(X[i_va], y[i_va]))
    oof[i_va] += model.predict(X[i_va])
```

- Source: stage-1 blend test probabilities. 7,455 rows cross (99.3% Dead).
- The validation fold and the OOF array never contain a pseudo-row; the estimate stays unbiased.
- Early stopping still watches only real labelled rows.
- Stage 2 = same trio, 10 folds $\times$ 2 seeds, retrained on the augmented folds.
- Tested and rejected: bands 0.05/0.97 (14,566 rows), and a second round from stage-2 predictions.

Part 3 · Section 3

Reproducibility

3

Reproducibility checklist

Determinism

- Fold seeds 42 / 2026 / 7; every model seeded with the fold seed; sklearn models seed 42.
- Tuned parameters hard-coded (no live Optuna run needed).
- Same `build_features()` in every script and in the notebook.

Environment

- Python 3.12.13 · LightGBM 4.7.0 · XGBoost 3.4.1 · CatBoost 1.2.10 · scikit-learn 1.9.0 · Optuna 4.9.0 · pandas 3.0.5 · NumPy 2.5.2
- One Apple M2 laptop, 8 cores, 8 GB RAM, CPU only (TabPFN test used the GPU).

Verification

- Notebook (17 cells) re-run in a clean kernel before submission.
- Output matches sub_014 on 36,000 / 36,000 rows; sub_016 = sub_014 + the one documented duplicate correction.
- Runtime $\approx$ 1.5 h end to end.
- `check_submission.py` asserts: same columns, same row count, same id order, no NaN, only labels present in the sample.
- No external data, no manual labels, no AutoML.

Traceability: where each headline number lives

Claim in Part 1	Where a reviewer can check it
OOF weighted F1 0.87996	notebook, final-ensemble cell; ledger row 17; <code>data/processed/oof_v4_blend.npy + oof_div2_*.npy</code>
Private 0.880410, public 0.875506	Kaggle submission log, file <code>sub_016_leakfix-of-014.csv</code>
Death rate 93.7% → 44.4% by year	notebook EDA cell (<code>groupby("year_of_diagnosis")</code>)
7,455 pseudo-labelled rows, 99.3% Dead	notebook stage-2 cell; <code>src/train_v4.py</code> run log
Diversity +0.0006 (split-half)	ledger row 17; <code>src/train_diverse2.py</code> OOF files
Threshold 0.575	notebook final cell (best_threshold) and per-fold medians in <code>src/train_v4.py</code>
Tuned LightGBM parameters	<code>data/processed/lgb_tuned.json</code> ; <code>src/tune.py</code>
283 duplicate test rows	ledger row 16; a 35-column merge of test against train
Optuna, distillation, per-year thresholds rejected	ledger rows 3–4, 22, and the insights file, each with the probe submission named

In one sentence

Every number on a slide is either a printed line of a seeded script or a leaderboard entry, and the file that produced it is named.

Thank you.

The repository, the notebook and the ledger are open. Pick any file.

Team NeurAps · Insight 2.0 — Datathon 2026 · private weighted F1
0.880410 (4th of 79)